

CONJURA · OPEN PROBLEM

Randomness Extraction from Unpredictable Random-Oracle Sources

A Leftover-Hash-Lemma Conjecture, Public Seed

Status. Statement: human-written, formalized in Lean (see `lean/Statement.lean`).

Category. Information-Theoretic Cryptography.

Setting. Fix nonempty finite sets $\mathcal{K}$ (seeds), $\mathcal{D}$ (inputs), $\mathcal{R}$ (outputs), and write $K := |\mathcal{K}|$, $D := |\mathcal{D}|$, $R := |\mathcal{R}|$. Let $\text{Fun}(\mathcal{K} \times \mathcal{D}, \mathcal{R})$ be the set of all functions $\mathcal{K} \times \mathcal{D} \rightarrow \mathcal{R}$, let $\text{SD}(\cdot, \cdot)$ denote statistical distance and $U_{\mathcal{R}}$ the uniform distribution on $\mathcal{R}$. The source S , the predictor P and the distinguisher D of Figure 1 are computationally unbounded and receive the *entire* function table of H as an explicit input. The seed sd is sampled only after S has terminated, and is given to D : that is what makes this the *public-seed* game.

Game Pred_P^S $H \xleftarrow{\\$} \text{Fun}(\mathcal{K} \times \mathcal{D}, \mathcal{R}); \quad (x, z) \xleftarrow{\\$} S(H); \quad x' \xleftarrow{\\$} P(H, z)$ return $(x = x')$
Game Ext-pub_D^S (public seed) $H \xleftarrow{\\$} \text{Fun}(\mathcal{K} \times \mathcal{D}, \mathcal{R}); \quad (x, z) \xleftarrow{\\$} S(H); \quad sd \xleftarrow{\\$} \mathcal{K}$ $y_0 \leftarrow H(sd, x); \quad y_1 \xleftarrow{\\$} \mathcal{R}; \quad b \xleftarrow{\\$} \{0, 1\}$ $b' \xleftarrow{\\$} D(H, sd, y_b, z); \quad \textbf{return } (b = b')$

Figure 1: Prediction game (top) and the public-seed extraction game (bottom). The first argument of S , P and D is the *full function table* of H ; the distinguisher additionally receives the seed sd .

Definition 1. Set $\text{Adv}_{\mathcal{K}, \mathcal{D}, \mathcal{R}, S}^{\text{pred}}(P) := \Pr[\text{Pred}_P^S \Rightarrow 1]$ and

$$\text{Adv}_{\mathcal{K}, \mathcal{D}, \mathcal{R}}^{\text{ext-pub}}(S, D) := 2 \Pr[\text{Ext-pub}_D^S \Rightarrow 1] - 1.$$

A source S is ϵ -unpredictable if $\text{Adv}_{\mathcal{K}, \mathcal{D}, \mathcal{R}, S}^{\text{pred}}(P) \leq \epsilon$ for every unbounded predictor P .

Conjecture 1 (Public seed). *There is a universal constant $c > 0$, independent of $\mathcal{K}$, $\mathcal{D}$, $\mathcal{R}$ and ϵ , such that for all nonempty finite $\mathcal{K}, \mathcal{D}, \mathcal{R}$, all $\epsilon \in (0, 1]$, every ϵ -unpredictable source S and every unbounded distinguisher D ,*

$$\text{Adv}_{\mathcal{K}, \mathcal{D}, \mathcal{R}}^{\text{ext-pub}}(S, D) \leq \delta_{\text{pub}}(\epsilon, K, D, R) := c \cdot \left(\sqrt{\epsilon R} + \sqrt{\frac{\log_2 D}{K}} \right).$$

Remark 1 (What the leftover hash lemma gives). Condition on the table and the auxiliary output, $(H, Z) = (h, z)$, and write $p_x := \Pr[X = x \mid h, z]$. Because sd is drawn only after S has terminated it is uniform on $\mathcal{K}$ and independent of X given (h, z) , so the distinguisher's task is to tell $(sd, h(sd, X))$ from uniform on $\mathcal{K} \times \mathcal{R}$: a strong-extractor question about the *fixed* family $\{h(k, \cdot)\}_{k \in \mathcal{K}}$, to which the leftover hash lemma applies in its almost-universal form. Writing $\text{CP} := \sum_x p_x^2$ for the source's collision probability and $\bar{\beta} := \frac{1}{K} \sum_{k \in \mathcal{K}} \Pr[h(k, X) = h(k, X') \mid X \neq X']$ for the family's mean collision probability on that source, it bounds the conditional advantage by

$$\frac{1}{2} \sqrt{R \cdot \text{CP} + \gamma}, \quad \gamma := R\bar{\beta} - 1,$$

where $\gamma \leq 0$ exactly when the family is universal on the source's support. Averaging over (H, Z) preserves the shape, $\sqrt{\cdot}$ being concave, and $\text{CP} \leq \max_x p_x$ turns the first summand into $\frac{1}{2} \sqrt{\epsilon R}$.

Two things follow, and between them they say what this conjecture asks for that the lemma does not supply. On the first summand the lemma already delivers the conjectured shape, with $c = \frac{1}{2}$. On the second it cannot. The family $\{h(k, \cdot)\}$ is not universal and cannot be made so: a universal family on D inputs needs a seed of $\Omega(\log D)$ bits, that is $K \geq D^{\Omega(1)}$, whereas the conjecture is of interest at $K \approx \log_2 D$ (Vadhan, *Pseudorandomness*, the remark following Theorem 6.18). So γ must be certified for a random table, uniformly over the supports S may pick *after* seeing H , and whatever certification is used it enters under the square root. Since $\bar{\beta}$ is a collision frequency read off the Kt table entries above a support of size t ,

and the estimate must hold uniformly over $\binom{D}{t}$ of them, it cannot be pinned closer than $\sqrt{\log D/K}$; so the lemma yields

$$\mathbf{Adv}_{\mathcal{K}, \mathcal{D}, \mathcal{R}}^{\text{ext-pub}}(S, D) \lesssim \frac{1}{2} \sqrt{\epsilon R} + o\left(\left(\frac{\log_2 D}{K}\right)^{1/4}\right),$$

a fourth root where δ_{pub} asks for a square root. In seed length that is $\log_2 K = \log_2 \log_2 D + 4 \log_2(1/\delta)$ against the conjectured $\log_2 \log_2 D + 2 \log_2(1/\delta)$, the latter being optimal for any extractor (Radhakrishnan and Ta-Shma, *SIAM J. Discrete Math.* 13(1), pp. 2–24, 2000).

The conjecture is therefore not a corollary of the leftover hash lemma, and its whole content is the second summand: the price of letting the source choose its support after seeing the entire table. The solution note in this folder closes that gap by never passing through a collision probability, bounding the L_1 distance directly instead.

The secret-seed companion. This note states one conjecture, the one the accompanying page states. There is a companion statement in the same setting, differing only in that D is called as $D(H, y_b, z)$ and never receives sd ; it conjectures the bound $c\sqrt{(\epsilon R + \log_2 D)/K}$, in which the seed's factor $1/K$ attaches to both summands. It is a genuinely different claim, not a restatement: the solution note in this folder proves the conjecture above and separately shows that transposing the secret-seed expression verbatim into the public-seed game gives a *false* statement. The two were originally posed together in a single write-up and were split so that each note states exactly the conjecture its page states.